SAP Customer Analysis: Royal Greenland

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Part 1: Review Royal Greenland customer story materials

Key Stakeholders:

1. ***SAP Project Team –*** Enterprise Software architects, solution consultants, and cloud engineer responsible for provisioning SAP Business Technology Platform (SAP BTP) environments, setting up secure APIs, and integrating core ERP and HANA databases with external applications.
2. ***SAP Ecosystem Partner (Trifork) –*** Digital design, mobile engineering, and implementation partner that leveraged the SAP BTP SDK for iOS to design user-friendly, mobile and tablet applications specifically tailored to harsh maritime environments and with offline-functionality.
3. ***Royal Greenland stakeholders –*** Executive management, landing site/ Harbor managers, back-office procurement and finance administrators, supply chain logistics planners, and sustainability compliance officers overseeing fish catch quotas and export certifications.
4. ***Fishermen & Harvesters –*** Primary producers operating in remote coastal waters who capture the initial catch data (such as, species, estimated weight, time, location) and rely on prompt, accurate landing confirmations and payments.
5. ***Regulatory & Certification Authorities –*** Marine stewardship bodies (Eg., MSC) Greenland fisheries license control (GFLK), and export customs authorities that demand traceable, audit-able and verified sustainable fishing quota records before export clearances are granted.

Royal Greenland's digital transformation goals:

- ***Automate Catch-to-procurement Workflow:*** Transition from manual, greasy paper landing forms to fully automated digital catch registration, eliminating processing backlogs and manual form filling at landing stations.
- ***Enable Real-Time Operational Visibility:*** Provide Harbor managers and supply chain planners with immediate data regarding fish volumes (estimated weight), species, timing and location of the catch.
- ***Optimize Traceability and Regulatory Compliance:*** Automate the compliance validation against national fishing quotas and global sustainability standards (MSC certification) to ensure the catch is ready for export.

- ***Accelerate Fisher Settlement Cycles:*** Enables instantaneous calculation and issuance of purchases orders (Pos) and receipts upon landing to improve trust, transparency and cash-flow for local fishers.
- ***Bridge connectivity Gaps in Remote Regions:*** Deploy resilient, offline-capable digital tools that ensure zero operational interruption even when vessels operate far outside cellular coverage.

Component Impacted:

The following digital transformation component is impacted: Business Process Transformation (alongside significant Organizational & Cultural Transformation).

Explanation & Examples:

The core operational process of how raw catch data moves from the ocean to enterprise ERP systems was fundamentally redesigned. Previously, fishers manually filled out physical paper tickets, hand-delivered them to harbor offices and waited while administrative clerks keyed that information into legacy systems for certification and purchase order generations. Under the new solution, this paper-based sequence was transformed into a digital, touch-driven workflow where catch is logged on a mobile app at sea, synchronized automatically at shore via SAP BTP, and processed directly into SAP ERP with automated purchase order generation. This fundamentally altered day-today work habits for both remote fishers and shore administrators.

Key metrics:

- ***70,000+ Purchase Orders Processed Digitally:*** Eliminated thousands of hours of manual paper data entry and paper archiving.
- ***200+ Active Fishermen Onboarded:*** High user adoption rate across diverse, distributed coastal communities with minimal training required due to intuitive UI design.
- ***Near 100% Reduction in Catch Registration Data Errors:*** Eliminated transcription errors, illegible handwriting issues and delayed quota reporting.
- ***Significant Cycle-time reduction in the Fisher Payouts:*** Reduced administrative settlements time from days/weeks down to near real-time receipts upon catch landing.

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Part 2: Understand SAP BTP capabilities

SAP BTP technology capabilities:

- ***Application development –*** For process automation, Developing a digital experience, specialized in visual low-code/ No-code programming and DevOps.
- ***Integration –*** Integrations such as Process integration, API-led integration, Event based integration and data integration.
- ***Data and Analytics –*** Setting up and managing of Databases, Data management, Data warehouses and analytics & planning
- ***Artificial Intelligence –*** Such as Prebuilt Ai models, intelligent processes, self-learning programs and data security & Decentral persistence.

Utilized SAP BTP capabilities to achieve the goals:

- ***Application Development –*** Utilized to develop custom, native iOS applications for fishers and web applications for harbor managers using the SAP BTP SDK for iOS, providing offline-first functionality.
- ***Integration –*** Utilized via SAP Integration Suite/ Cloud Integration to securely synchronize data captured across edge mobile applications with Royal Greenland's core SAP ERP and SAP Integrated Business Planning (IBP) back-end systems.
- ***Data & Analytics –*** Utilized to ingest, process and consolidate catch quantities in real time, giving supply chain operators instant visibility into factory inventories, quota limits and processing yields.
- ***AI & Automation –*** Utilized to automate digital document generation (e.g., auto-generating ~70,000 digital purchase orders and compliance documentation) without requiring human intervention at the terminal.

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Part 3: Review the end-to-end SAP Solution

Deployed Architecture:

The Royal Greenland architecture is deployed as a **Hybrid Cloud Architecture**, blending cloud-native edge services with enterprise transactional back-ends:

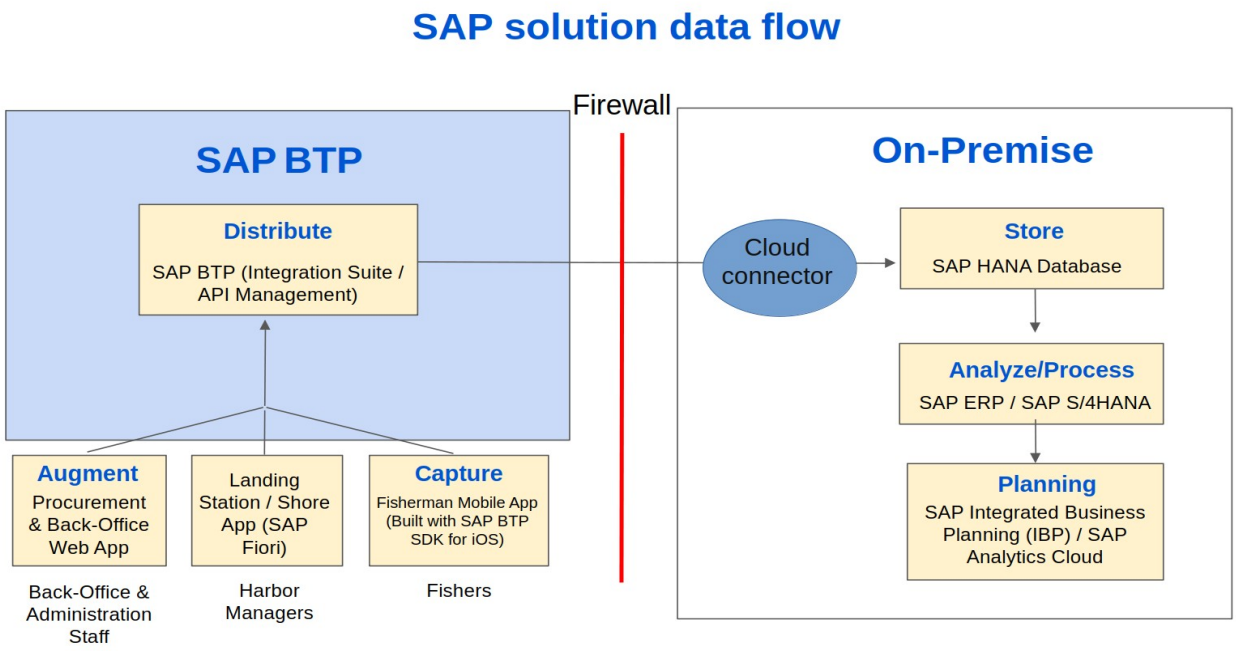
- ***Front-End (User-Facing Layer):*** Consists of three tailored applications:
  1. ***Fisherman App –*** Native iOS mobile application built with SAP BTP SDK for iOS, designed with offline local storage for logging catch at sea.
  2. ***Landing Station App –*** Web/Tablet application (SAP Fiori) used by harbor inspectors to weigh, grade, verify and augment catch records.
  3. ***Administrative/ Procurement Portal –*** Used by back-office staff for finance, supplier management and export certification.
- ***Middle-ware & Integration (SAP BTP Layer):*** Serves as the central cloud hub. It manages user authentication, offline data sync queuing, API management (OData/REST) and business workflow routing.
- ***Back-end (Core System & Storage):*** Built on SAP ERP (S/4HANA/ECC), SAP HANA Database and SAP Integrated Business Planning (IBP). This layer stores master records, processes automated purchase orders, deduct volumes from regulatory quotas and feeds downstream global supply chain logistics.

Points to Consider:

The project team will need to consider the following when building the end-to-end solution:

- ***Computing Models –*** Cloud-first for front-end access, APIs and scalable mobile workloads on SAP BTP, integrated with enterprise compute and storage running in private cloud/ on-premise environments for transactional ERP records.
- ***Architecture –*** Hybrid cloud model ensuring seamless bi-directional integration between external public internet/ mobile devices and protected internal corporate systems.
- ***Operating Systems and Platforms –*** iOS for mobile fishermen devices, web-standard cross-platform browsers for shore terminals (SAP Fiori) and SAP BTP runtime environments as the unifying platforms.
- ***Application Development –*** Responsive, offline-first design patterns with local data caching to accommodate low or non-existent marine bandwidth, paired with high-contrast UI for arctic sunlight and weather conditions.
- ***Programming language –*** Swift for native iOS application, JavaScript/Type-Script (SAPUI5/HTML5) for web applications and ABAP/Core Data Services (CDS) for core back-end ERP business logic.
- ***Data Analytics –*** Requirements for capturing catch location, timestamp, weight, temperature and species to track real-time quota consumption.

Data flow through the solution:



Areas Impacted:

The technology areas impacted by further development include:

- ***E-Commerce and Inventory Management (e.g., SAP Commerce Cloud) –*** because implementing a digital platform for buying fishing equipment requires integrating a customer-facing product catalog with real-time warehouse inventory and supply chain systems. It also involves introducing digital payment gateways or automated deduction-at-source workflows that debit transactions directly from the fisher's catch earnings in SAP ERP.
- ***Geo-spatial Processing, IoT Telemetry and Emergency Communications –*** because utilizing mobile applications to locate boats lost at sea necessitates continuous GPS coordinate capture, telemetry caching and resilient satellite or cellular data transmission. It also requires establishing secure, real-time API integrations with maritime rescue authorities and coastal emergency command centers to share vessel locations during distress events.
- ***Data Analytics and Predictive Intelligence (e.g., SAP Analytics Cloud / Embedded Micro-services) –*** because delivering financial literacy, budgeting and saving tools requires aggregating historical catch volume, market pricing trends and payout records. These data sets must be transformed into intuitive mobile dashboards that project seasonal income, display operational cost trends and help fishers set automated saving goals.
- ***Information Security, Identity Management and Data Privacy –*** because expanding into commercial purchases, vessel tracking and personal finance introduces sensitive geo-location, payment and financial data. This mandates strict compliance with data protection standards, end-to-end encryption for data in transit and at rest and robust role-based access control (RBAC) managed via SAP BTP.

Part 4: Assess how the SAP solution supports digital transformation

SAP BTP Support:

SAP BTP supports Royal Greenland's digital transformation goals by:

- ***Decoupling Front-End Innovation from Core ERP Transactions –*** Providing an agile, cloud-based platform-as-a-service (PaaS) foundation that allowed Royal Greenland to modernize edge operations (using the SAP BTP SDK for iOS) without disrupting or re-architecting its underlying mission-critical SAP ERP and HANA databases systems.
- ***Establishing Seamless End-to-end Enterprise Integration –*** Leveraging the SAP Integration Suite to automate bi-directional data synchronization between remote offline mobile clients, landing station web applications and central supply chain management solutions (SAP Integrated Business Planning). This eliminated administrative latency and replaced error-prone manual paper entry with automated digital purchase orders and instant receipts.
- ***Delivering Real-Time Operational Visibility & Decision Support –*** Consolidating real-time landing data into centralized analytical repositories, enabling operational leaders to forecast factory processing capacity, balance supply chain logistics and optimize fleet resource distribution dynamically.
- ***Creating a Scalable Foundation for Continuous Innovation –*** Offering an extensible cloud ecosystem capable of supporting future value-added initiatives – such as e-commerce gear procurement, IoT maritime emergency telemetry and embedded financial wellness micro-services without requiring architectural redevelopment.

SAP Solution's Contributions:

People	***Planet***	***Profit***	***Purpose***
• Eliminates cumbersome, error-prone paperwork in harsh Arctic weather conditions. • Delivers instantaneous catch confirmations, digital receipts	• Enforces automated, real-time monitoring against national quotas and Marine Stewardship Council (MSC) standards. • Reduces the risk of overfishing, preserving marine	• Automates manual administrative processes, digitizing over 70,000 purchase orders and drastically lowering back-office labor overhead. • Minimized data-	• Supports the economic resilience and independence of remote coastal communities in Greenland. • Connects independent artisanal fishers directly into modern,

and transparent settlements statements to mobile devices • Features an intuitive user experience tailored for low learning curves and zero disruption to daily fishing tasks	biodiversity in sensitive Arctic waters. • Eliminated physical paper waste, physical record storage overhead and courier transport footprints across Isolated coastal hubs.	entry errors, audit penalties and catch settlement discrepancies • Enhance factory throughput, supply chain visibility and overall inventory planning accuracy	transparent global supply chains. • Advances responsible, verifiable environmental stewardship across the broader seafood industry.

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